

JEONGHWAN ‘JAY’ LEE

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SUMMARY

Ph.D. in Mechanical Engineering with expertise in **human biomechanics, robotics, rehabilitation engineering, and data-driven modeling**. Over 5 years of human-centered research experience spanning experimental design, quantitative analysis, statistical and machine learning modeling, and multimodal sensor data analysis (IMU, EMG, load cells, motion capture, vision sensors etc.). 3 years of industry experience leading **end-to-end robot perception stack development, human-in-the-loop MLOps pipelines, and test & validation task forces** for successful production deployment. Proven track record of delivering production-grade solutions in high-growth startup and scale-up environments.

CURRENT AFFILIATION

Massachusetts Institute of Technology

Cambridge, MA

Postdoctoral Associate at the Department of Brain and Cognitive Science

Apr 2025 – Present

- **Developing quantitative human-likeness metrics** grounded in motor behavior to evaluate AI-generated human motion.
- **Modeling high-DOF musculoskeletal predictive simulations of human motion** by integrating modern generative AI with computational neuromechanics.
- **Collaborating interdisciplinary research projects:** i) data-driven functional and cognitive assessments of the cutaneous mechanoneuronal interface for neuroprosthesis (with MIT Media Lab); ii) exploration of how humans and generative AI compress information into flexible, efficient representations in vision and language (with NYU).

INDUSTRY EXPERIENCE

Contoro Robotics

Austin, TX

Senior Robotics Engineer – Perception

Sept 2022 – Mar 2025

- **Led the development of a ROS-integrated, multi-functional robot perception pipeline** with object detection, image segmentation, 6D pose estimation, optical character recognition, and teleoperation feedback, enhancing reliability and flexibility in real-world use.
- **Designed a cloud-based human-in-the-loop segmentation and online data re-labeling workflow** combining instance and zero-shot segmentation, achieving 2% operational failures and improving model robustness through continuous data annotation.
- **Developed an automated, multi-step hand-eye calibration methodology** for a multi-camera setup (eye-in-hand, eye-to-hand, and stereo), achieving localization accuracy within 0.5% of working distance.
- **Managed a test and validation task force** that integrated DevOps pipelines with multi-scale physical mockups, successfully narrowing the sim2real gap and securing multiple customer deployments.
- **Led technical demonstrations** that played a pivotal role in securing \$4.7 million in Seed and \$12 million in Series A funding.

Harmonic Bionics, Inc. (now Bioness Medical)

Austin, TX

System Validation Engineer Intern

May 2019 – Aug 2019

- **Developed C/C++ EtherCAT library** for motion controllers and sensor interfaces.
- **Built dual-motor haptic demo kits** for exhibitions, demonstrating advanced motion control capabilities.

RESEARCH EXPERIENCE

University of Texas at Austin

Austin, TX

Graduate Research Assistant

Aug 2017 – Aug 2022

- **Led three data-driven biomechanics and wearable robot research initiatives**—low-cost gait trainer design optimization, kinematic biomarkers identification for physical state-aware exoskeletal assistance, and post-stroke gait characterization—culminating in four first-authored publications.
- **Supervised two master’s students in biomechanics research** on gait biomechanics related to cognitive decline and exoskeleton design, resulting in two co-authored publications.
- **Managed the end-to-end research process:** study conceptualization, securing funding, data collection/analysis (kinematics, dynamics and EMG), interpretation, visualization, publication authorship, and presentation.

- **Collaborated with a multidisciplinary team** of neuroscientists and clinicians to drive innovative solutions.

Korea Institute of Science and Technology (KIST)

Seoul, South Korea

Research Assistant

Mar 2017 – July 2017

- **Devised a non-invasive, patient-specific surgical tool navigation method** for orbital reconstructive surgery, achieving up to 50% improvement in registration and tool tracking accuracy using a 3D-printed phantom model.

Seoul National University

Seoul, South Korea

Graduate Research Assistant

Mar 2016 – Feb 2017

- **Developed a needle steering scheme** using pivoted super-elastic Nitinol for an MR-guided breast needle intervention robot, enhancing insertion angle control with a targeting error of less than 5mm.
- **Designed a vehicle door and driver's seat mockup** with ten adjustable parameters, enabling controlled ingress/egress experiments.

The University of Texas Health Science Center (UTHealth)

Houston, TX

Research Assistant

Sept 2014 – Feb 2016

- **Prototyped and tested hysteresis** of a 7-DOF dual-segmented cable-driven continuum robotic manipulator for single-port surgery using a motorized testbed.
- **Contributed to transitioning a lab innovation into a commercial venture**, supporting initial product development and establishing scalable engineering processes (now Endoquest Robotics).

EDUCATION

Ph.D. in Mechanical Engineering , University of Texas at Austin, TX	2022
M.S. in Mechanical Engineering , Seoul National University, Seoul, South Korea	2017
B.S. in Mechanical Engineering , Hanyang University, Seoul, South Korea	2013

SKILLS

Programming:	Python, C/C++, MATLAB, Bash/Shell
Statistics / ML Platform:	SciPy, R, PyTorch, TensorFlow
DevOps / SysAdmin:	Docker, Git, Linux/Unix
Robotics Middleware:	Robot Operation System (ROS / ROS2)
Simulation Tools:	MuJoCo, Isaac Gym/Lab, OpenSim, Simulink, Simscape, ROS Gazebo
Design Tools:	SolidWorks, Eagle

CERTIFICATES

Data Science and Applied Machine Learning	May 2021
Scalable Machine Learning: Methods and Tools	May 2021
The University of Texas at Austin, Department of Statistics and Data Sciences	

SELECTED PEER REVIEWED ARTICLES (3 OUT OF 12) – [GOOGLE SCHOLAR](#)

*Selected works focus on data-driven approaches, biomechanics, and wearable sensors for developing personalized solutions in post-stroke rehabilitation.

1. **Lee, J.**, Seamon, B. A., Lee, R. K., Kautz, S. A., Neptune, R. R., & Sulzer, J. S. (2025). Post-stroke Stiff-Knee gait: are there different types or different severity levels?. *Journal of NeuroEngineering and Rehabilitation*, 22(1), 36.
2. **Lee, J.**, Akbas, T., & Sulzer, J. (2023). Hip and knee joint kinematics predict quadriceps hyperreflexia in people with post-stroke Stiff-Knee gait. *Annals of Biomedical Engineering*, 51(9), 1965-1974.
3. **Lee, J.**, Li, L., Shin, S. Y., Deshpande, A. D., & Sulzer, J. (2021). Kinematic comparison of single degree-of-freedom robotic gait trainers. *Mechanism and Machine Theory*, 159, 104258.

PATENTS

1. Varghese, R. J.; Watts, T.; Moser, N.; Hingwe, A.; Kim, M.; **Lee, J.**; Yun, Y. Articulated Multi-Zone Robot End Effector Systems and Methods for High Moment Loads. US20250282064A1, pending.
2. Kim, Y., **Lee, J.**, Park, S., Park, S.M., Cho, H., Kim, L., Noh, G., Lee, J.W., Lee, B.H., (2020). Automatic bending apparatus of plate for surgery, Republic of Korea (KR) Patent, No. 1021566940000.